Prompt 1 — Create the basic lab application

Create a simple educational web application for an undergraduate AIML laboratory.

Title:

Discrete Probability Distributions

The application will eventually contain:

1. Binomial Distribution

2. Poisson Distribution

3. Geometric Distribution

For now, implement only the basic application structure and the Binomial Distribution module.

Keep the UI simple and clean. Do not add unnecessary features.

Create a navigation/header with:

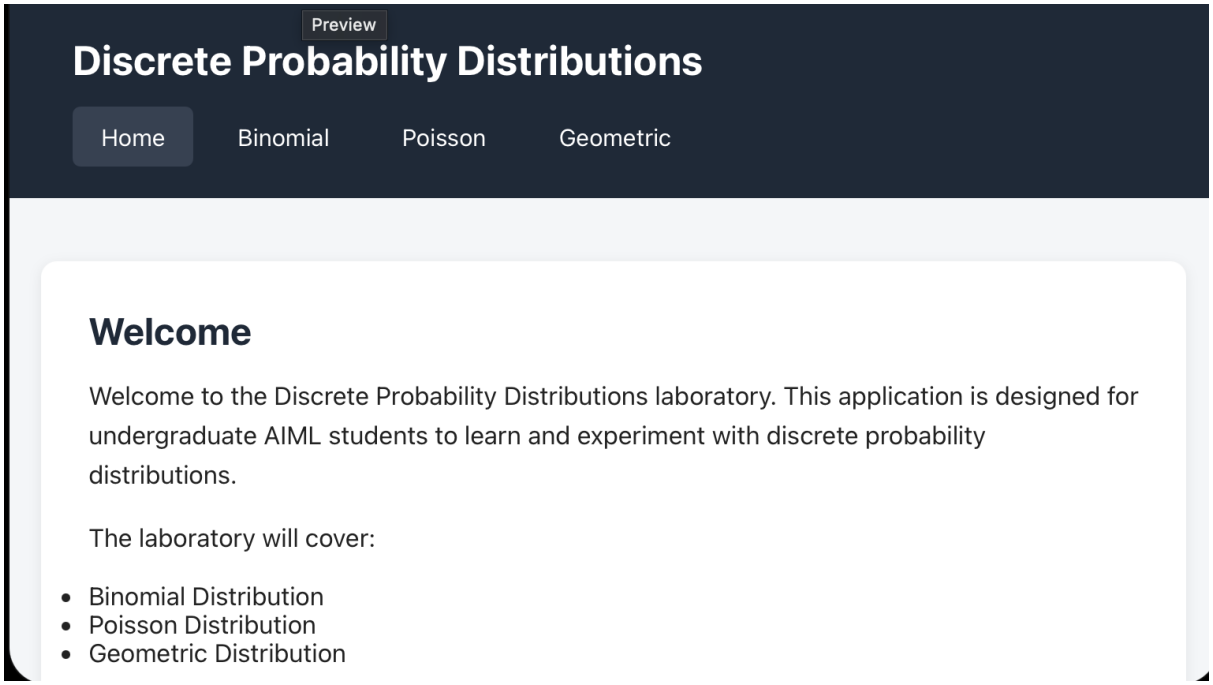
- Home

- Binomial

- Poisson

- Geometric

Only the Binomial page needs to be functional at this stage.



Binomial Distribution

Prompt 2 — Add Binomial inputs

Implement the Binomial Distribution section.

Add two input controls:

- Number of trials n

- Probability of success p

Set default values:

n = 20

p = 0.35

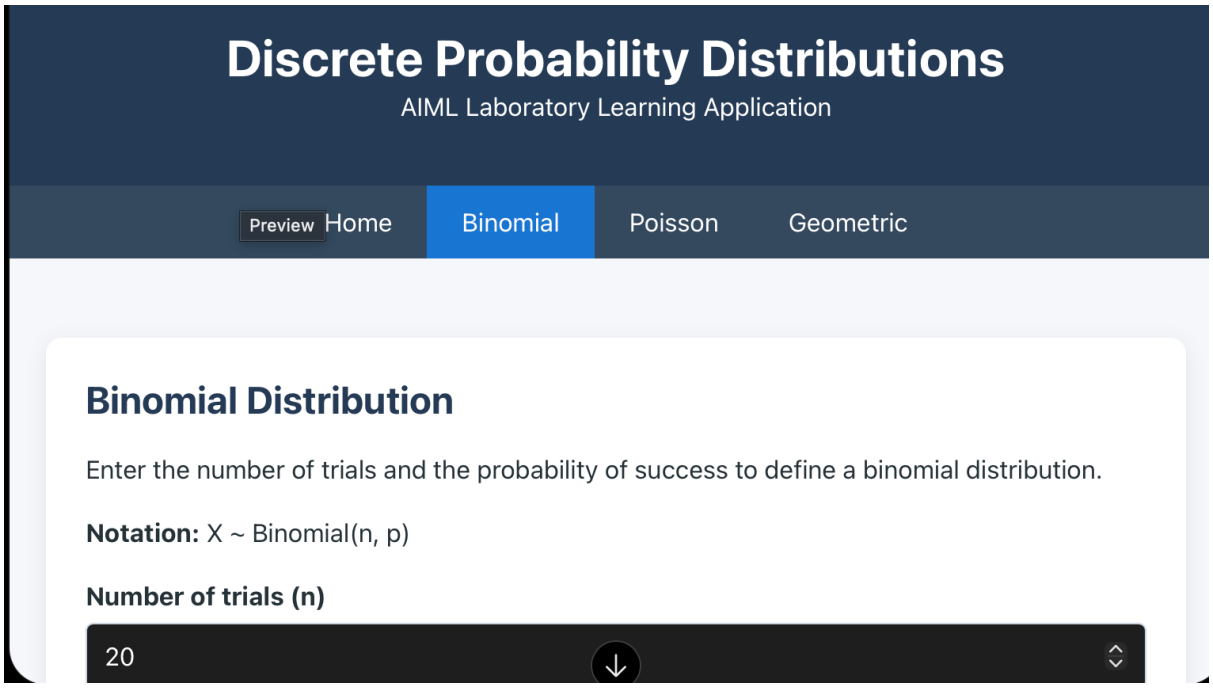
Validate the inputs:

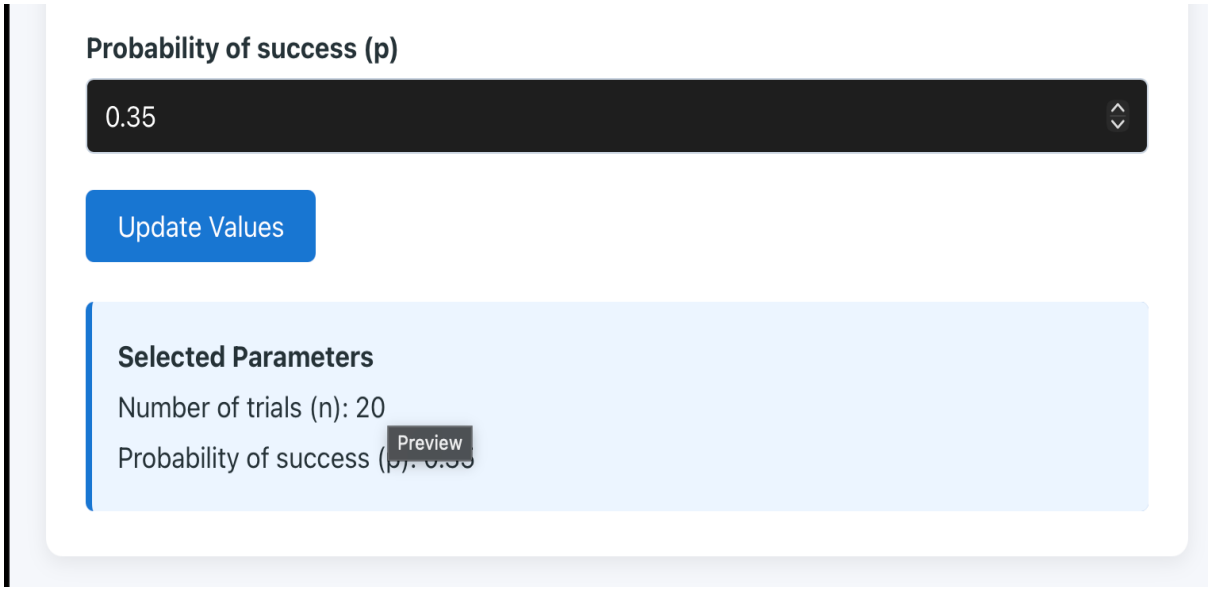
- n must be a positive integer

- p must be between 0 and 1

Display the selected values clearly.

Do not add simulation yet.





Prompt 3 — Calculate Binomial statistics

Add the Binomial probability calculations using the selected n and p.

Calculate and display:

- Mean

- Variance

- Standard deviation

- Second moment $E[X^2]$

- Skewness

Use the following formulas:

Mean:

np

Variance:

np(1-p)

Second moment:

Variance + Mean²

Skewness:

(1-2p) / sqrt(np(1-p))

Display the results in simple statistic cards.

For the default values n=20 and p=0.35, verify that the mean is 7 and variance is 4.55.

Binomial Distribution

The binomial distribution describes the number of successes in a fixed number of independent trials.

Notation: X ~ Binomial(n, p)

Number of trials (n)

20

Probability of success (p)

0.35 Preview

Selected values: n = 20, p = 0.35

Binomial Statistics

The following statistics are calculated using the selected values of n and p.

Mean

7.0000

Mean = np

Variance

4.5500

Variance = np(1-p)

Standard Deviation

2.1331

SD = √Variance

Second Moment E[X²]

53.5500

E[X²] = Variance + Mean²

Skewness

0.1406

(1 - 2p) / Preview

Prompt 4 — Add PMF

Add a Binomial PMF calculator.

For:

X ~ Binomial(n,p)

calculate:

P(X = k)

Allow the student to enter k.

Validate that:

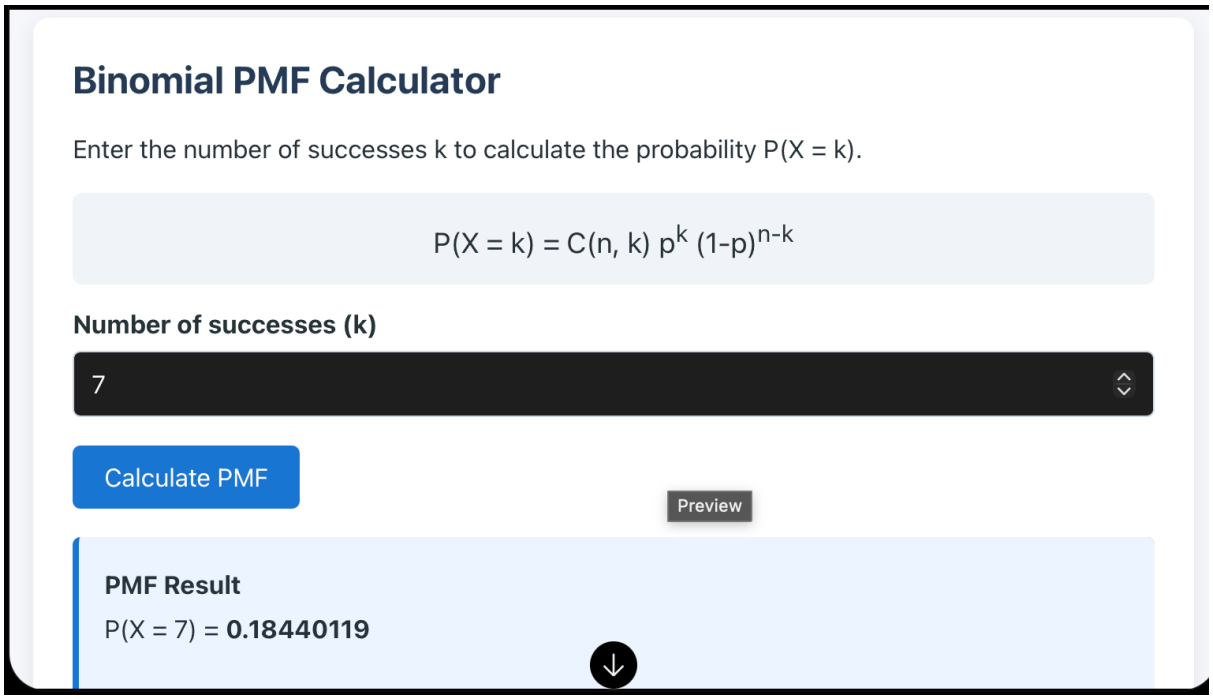
0 <= k <= n

Display the probability clearly.

Also display the mathematical expression:

P(X=k) = C(n,k) p^k (1-p)^(n-k)

Explain in one simple sentence what the result means.





Prompt 5 — Add CDF

Add a Binomial CDF calculator.

Allow the student to enter k.

Calculate:

P(X <= k)

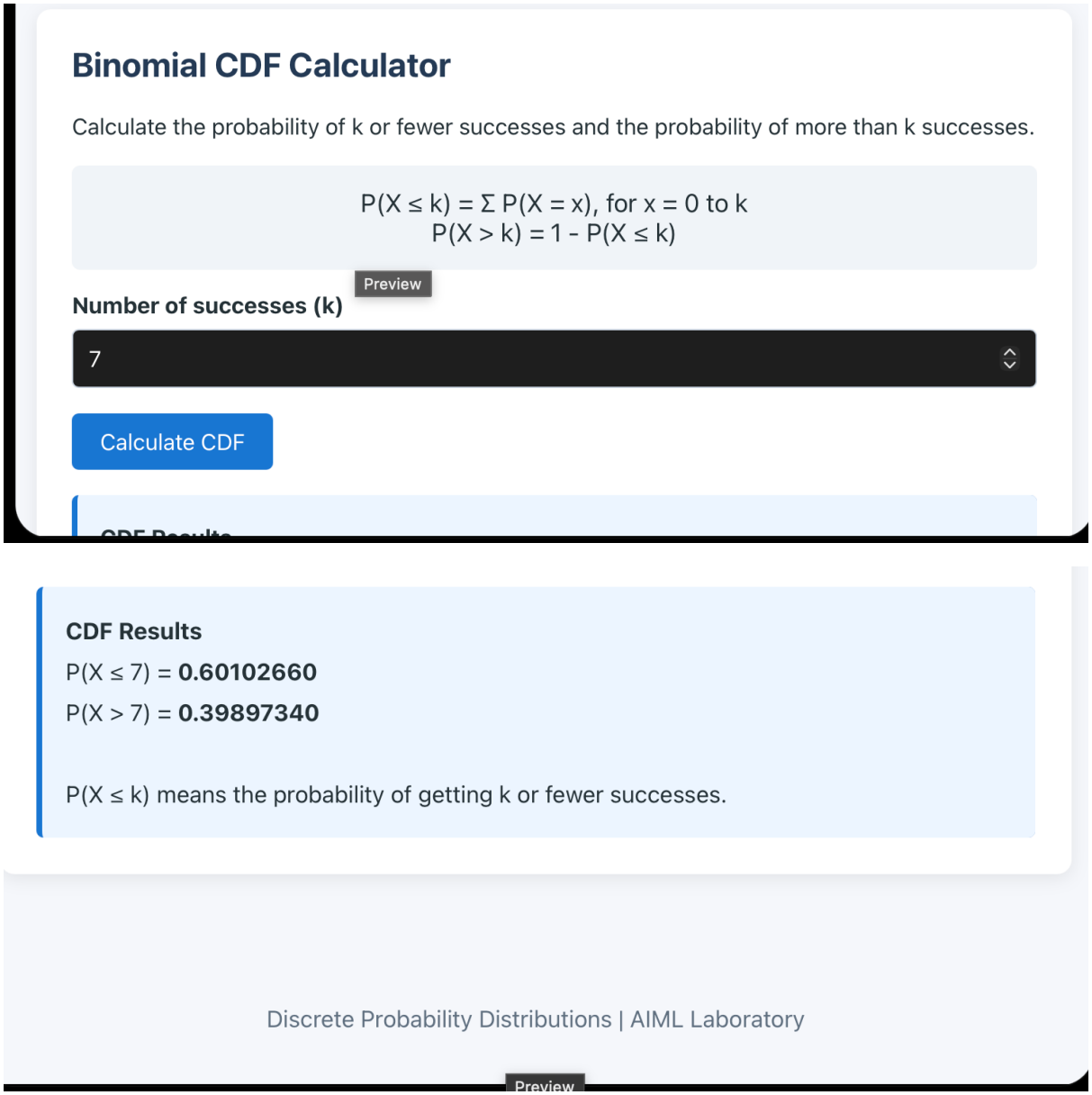
Also calculate:

P(X > k) = 1 - P(X <= k)

Display both results.

Explain:

“P(X <= k) means the probability of getting k or fewer successes.”



Prompt 6 — Add Binomial PMF graph

Add a bar chart showing the Binomial PMF.

X-axis:

Number of successes X

Y-axis:

Probability P(X)

Plot values for:

X = 0, 1, 2, ..., n

Use the current values of n and p.

Add a vertical dashed reference line at the mean.

Add a tooltip showing:

X = value

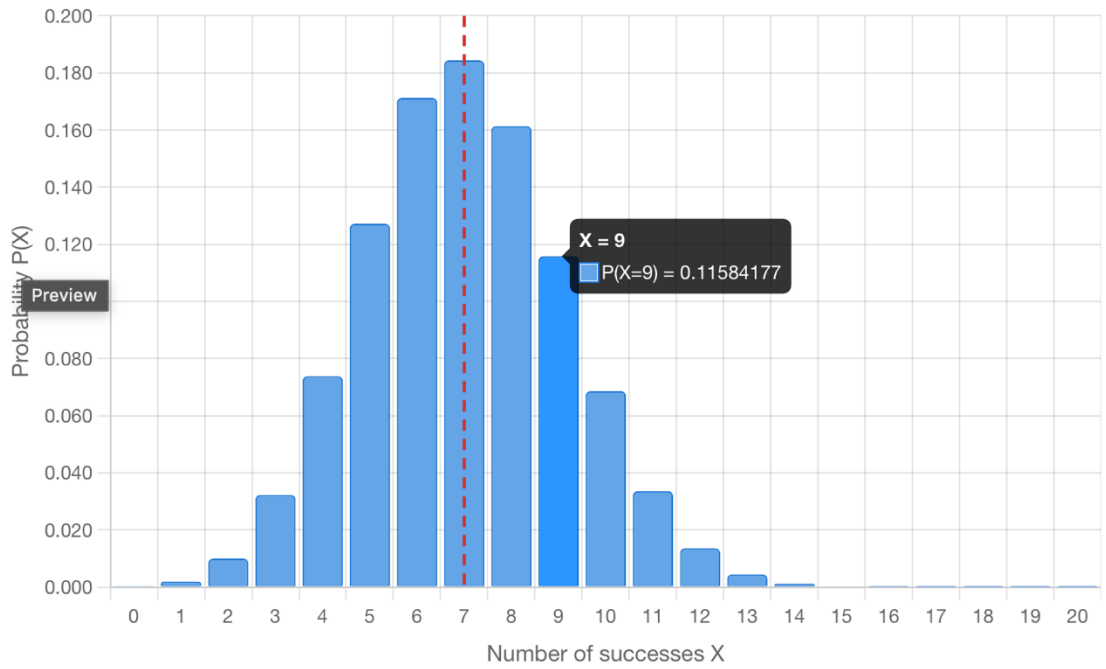
P(X=value) = probability

Update the graph automatically whenever n or p changes.

Preview

Bino PMF Graph

This bar chart shows the probability of each possible number of successes, from X = 0 to X = n. The dashed vertical line represents the mean (np).



X-axis: Number of successes X

Y-axis: Probability P(X)

Hover over any bar to see X and P(X=value).

Prompt 7:

Add a Poisson Distribution module to my existing “Discrete Probability Distributions” HTML application.

Keep the existing Binomial module fully functional.

Keep the same navigation, design, cards, buttons, and responsive layout.

Add Poisson to the navigation bar.

Set the default rate parameter to $\lambda = 4$.

λ must be greater than 0. Display an error message for invalid inputs.

Add Input Parameters, Statistics, PMF Calculation, CDF Calculation, and a PMF bar chart.

Calculate Mean = λ, Variance = λ, and Standard Deviation = $\sqrt{\lambda}$.

Use PMF: $P(X=k) = (e^{-\lambda} \times \lambda^k) / k!$, where k is a non-negative integer.

Use CDF: $P(X \leq k) = \Sigma\, P(X=i)$, for $i = 0$ to k.

Also calculate $P(X>k) = 1 - P(X \leq k)$.

Set the default value of k to 4.

Display the results in cards matching the existing Binomial module.

Use purple bars and a vertical dashed red line at the mean in the graph.

When λ changes and the Update button is clicked, automatically update the statistics and graph.

Validate inputs and display appropriate error messages.

Write the actual HTML, CSS, and JavaScript in my existing file. Do not create a separate application or change the existing Binomial module.

Return the complete updated HTML file.

Poisson Distribution

Calculate probabilities for event counts in a fixed interval.

Poisson · Purple

Input Parameters

Set the average number of events per interval.

Rate parameter (λ)

4

Preview

Greater than 0 and at most 100

Update Distribution

Poisson distribution updated.

Statistics

Mean (λ)

4

Variance (λ)

4

Standard Deviation (√λ)

2

PMF Calculation

$P(X = k) = e^{-\lambda} \lambda^k / k!$

Number of events (k)

4

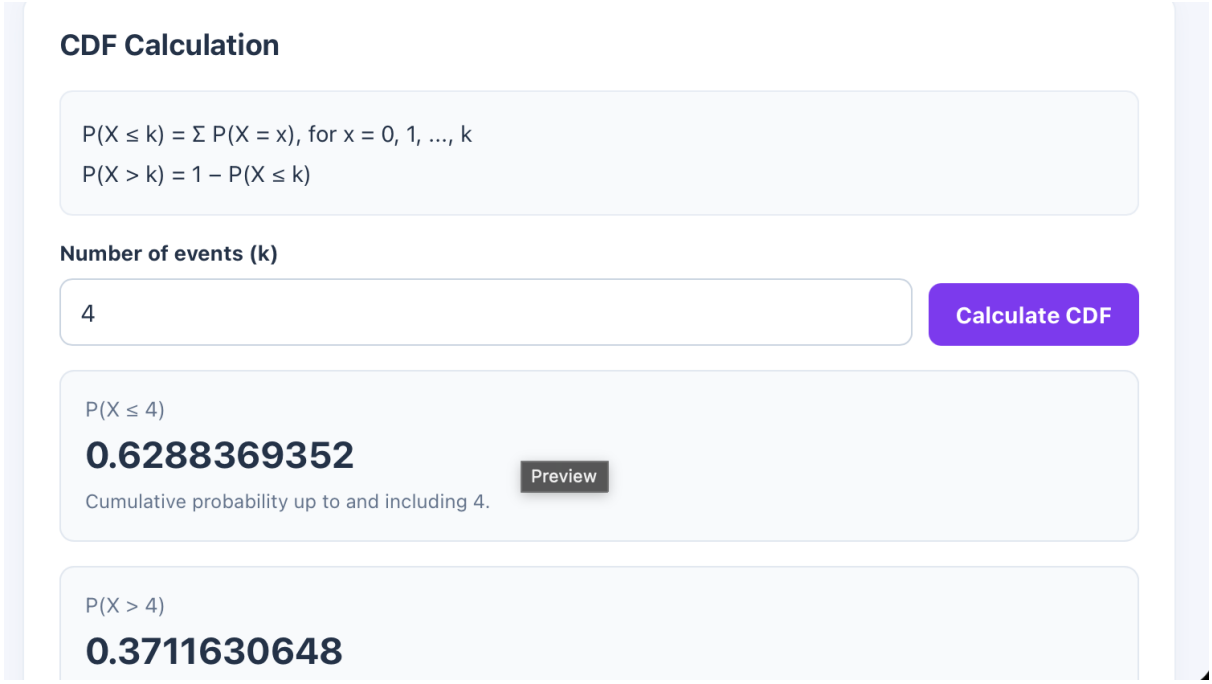
Calculate PMF

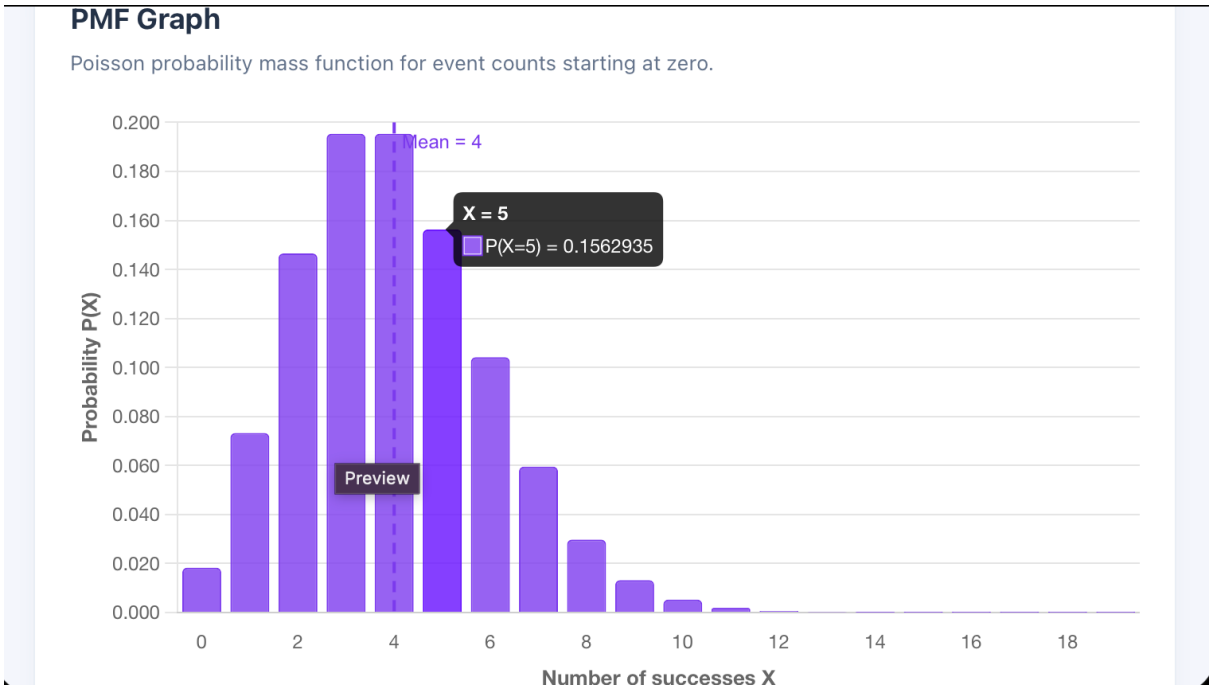
P(X = 4)

0.1953668148

Preview

Probability of exactly 4 events when





Prompt 8:

Add a Geometric Distribution module to my existing “Discrete Probability Distributions” HTML application.

Keep the existing Binomial and Poisson modules fully functional.

Keep the same navigation, design, cards, buttons, and responsive layout.

Add Geometric to the navigation bar.

Set the default probability of success to p = 0.35.

Use the convention X = trial number of the first success, starting from 1.

Add Input Parameters, Statistics, PMF Calculation, CDF Calculation, and a PMF bar chart.

Calculate Mean = 1/p, Variance = (1-p)/p², and Standard Deviation = √(1-p)/p.

Use PMF: P(X=k) = (1-p)^(k-1) × p.

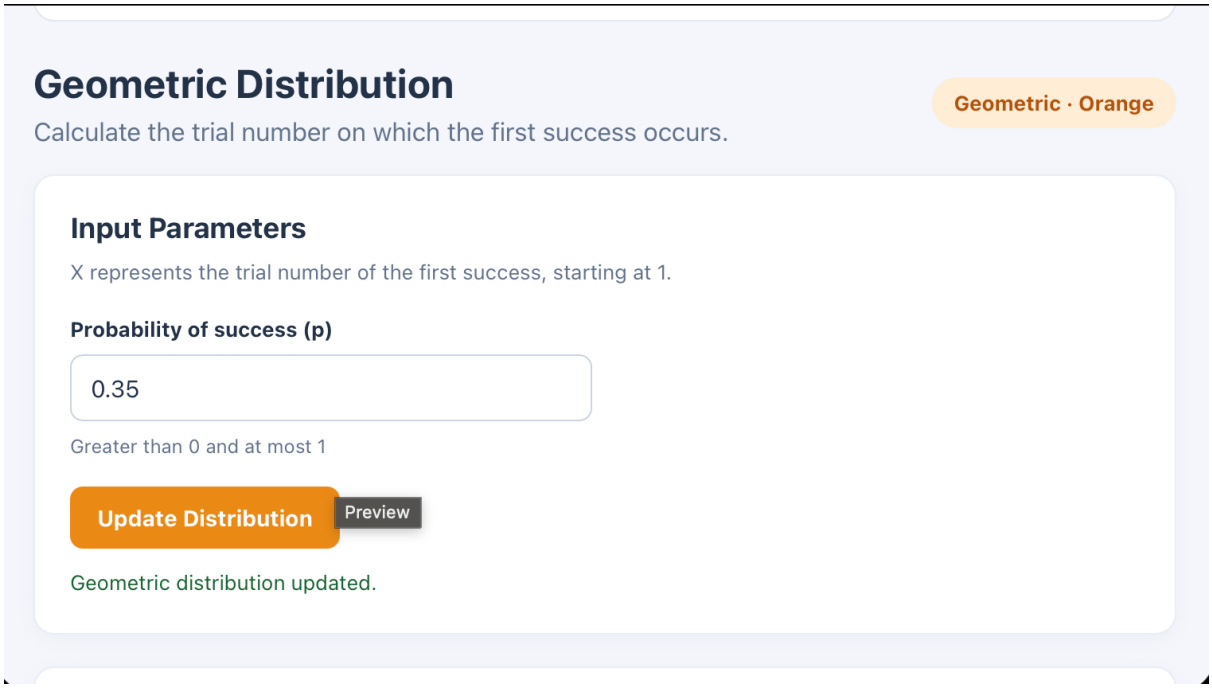
Use CDF: P(X≤k) = 1-(1-p)^k.

Validate inputs and display error messages.

Use orange bars and a vertical dashed red line at the mean in the graph.

Write the actual HTML, CSS, and JavaScript in my existing file. Do not create a separate application or change the existing modules.

Return the complete updated HTML file.



Preview

Statistics

Mean (1/p)

2.8571

Variance

5.3061

Standard Deviation

2.3035

PMF Calculation

$P(X = k) = (1 - p)^{k-1} p, k = 1, 2, 3, \ldots$

Trial number (k)

3

Calculate PMF

$P(X = 3)$

0.147875

Preview

Probability that the first success occurs on trial 3.

CDF Calculation

$P(X \leq k) = 1 - (1 - p)^k$

$P(X > k) = (1 - p)^k$

Trial number (k)

3

Calculate CDF

$P(X \leq 3)$ Preview

0.725375

Cumulative probability up to and including 3.

$P(X > 3)$

0.274625

Geometric probability mass function with trial numbers beginning at 1.

